

FORM TP 22244



TEST CODE **000574**

MAY/JUNE 2002

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

MATHEMATICS

UNIT 2 – PAPER 02

$2\frac{1}{2}$ hours

05 JUNE 2002 (p.m.)

This examination paper consists of **THREE** sections: Module 2.1, Module 2.2 and Module 2.3.

Each section consists of 2 questions.

The maximum mark for each section is 50.

The maximum mark for this examination is 150.

This examination consists of 5 pages.

INSTRUCTIONS TO CANDIDATES

1. **DO NOT** open this examination paper until instructed to do so.
2. Answer **ALL** questions from the **THREE** sections.
3. Unless otherwise stated in the question, all numerical answers **MUST** be given exactly **OR** to three significant figures as appropriate.

Examination Materials

Mathematical formulae and tables

Electronic calculator

Graph paper

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Section A (Module 2.1)

Answer BOTH questions.

1. (a) (i) Solve the simultaneous equations for $x > 0, y > 0$

$$xy = 4$$

$$2 \ln x = \ln 2 + \ln y.$$

[6 marks]

- (ii) Show that $\log_x y = \frac{1}{\log_y x}$ for x, y real and positive.

Hence, if $\log_x y = 2 + 3 \log_y x$, find y as functions of x .

[8 marks]

- (b) Find $\frac{dy}{dx}$ when

(i) $y = x^2 \tan 3x$

[4 marks]

(ii) $y = \frac{t}{1+t^2}, x = 3t^2.$

[7 marks]

Total 25 marks

2. (a) Find real constants A, B and C such that

$$\frac{3x^2 + 4x + 1}{(x+2)(x^2+1)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+1}.$$

Hence, evaluate

$$\int \frac{3x^2 + 4x + 1}{(x+2)(x^2+1)} dx.$$

[13 marks]

- (b) The rate at which atoms in a mass of radioactive material are disintegrating is proportional to n , the number of atoms present at any time, t , measured in days. Initially the number of atoms is m .

- (i) Form and solve the differential equation which represents this data.

[8 marks]

- (ii) Given that half of the original mass disintegrates in 76 days, evaluate the constant of proportion in the differential equation.

[4 marks]

Total 25 marks

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Section B (Module 2.2)

Answer BOTH questions.

3. (a) A sequence, $\{u_n\}$, is defined by $u_{n+1}u_n = 2^n$, $n \geq 1$.

Prove that $u_{n+2} = 2 u_n$.

[4 marks]

- (b) Express $\frac{1}{n}\sqrt{2n^2 + 3}$ in the form $\sqrt{p + \frac{q}{n^2}}$ where $p, q \in \mathbb{R}$.

Hence, show that $\lim_{n \rightarrow \infty} \frac{2n + 3}{\sqrt{2n^2 + 3}} = \sqrt{2}$.

[8 marks]

- (c) Given the series

$$\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots$$

- (i) obtain the n^{th} term of the series

[3 marks]

- (ii) find the sum of the first n terms of the series

[8 marks]

- (iii) find the sum to infinity if it exists.

[2 marks]

Total 25 marks

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4. The function f is given by $f(x) = x^4 - 4x + 1$.

Show that

- (a) when $x > 1$, f is strictly increasing [5 marks]
- (b) $f(x) = 0$ has a root in each of the intervals $[0, 1]$ and $[1, 2]$ [9 marks]
- (c) $f(x) = 0$ has no other roots in the intervals $[0, 2]$ [5 marks]
- (d) if x_1 is an approximation to the root of $f(x) = 0$ in $[1, 2]$, the Newton-Raphson method gives a second approximation
- $$x_2 = \frac{3x_1^4 - 1}{4(x_1^3 - 1)}$$
- in $[1, 2]$. [6 marks]

Total 25 marks

Section C (Module 2.3)

Answer BOTH questions.

5. (a) A manufacturer of computers is supplied with a particular computer microchip, called MC-40, from three suppliers, Halls Electronics, Smith Sales, and Crawford Sales and Supplies. A small batch of the chips supplied is defective. The information is summarised in the table below.

Supplier of Microchip, MC-40	% Supplied	% Defective
Halls Electronics	30	3
Smith Sales	20	5
Crawford Sales and Supplies	50	4

When the MC-40 chips arrive at the manufacturer, they are carefully stored in a particular container and not inspected, neither is the supplier identified.

- (i) A worker is asked to select ONE chip from the container for installation in a computer. Draw an appropriate tree diagram to represent this selection process. [8 marks]

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- (ii) What is the probability that
- a) it was supplied by Halls Electronics? [1 mark]
 - b) it was not supplied by Crawford Sales and Supplies? [1 mark]
 - c) it was supplied by Smith Sales and it was good? [1 mark]
 - d) it was defective? [4 marks]
 - e) it will work effectively in the computer? [4 marks]

- (b) A biology examination includes 4 True or False questions. The probability of a student guessing the correct answer to the first question is $\frac{1}{2}$. Likewise, the probability of a student guessing correctly each of the remaining questions is $\frac{1}{2}$. Use the probability model

$$P(r) = \frac{n!}{r!(n-r)!} \cdot p^r q^{n-r}$$

where n is the number of questions
 r is the number of observed successes
 p is the probability of guessing correctly
 q is the probability of guessing incorrectly,

to answer the questions below.

What is the probability of a student

- (i) guessing at LEAST ONE of the four questions correctly? [3 marks]
- (ii) guessing EXACTLY ONE of the four questions correctly? [3 marks]

Total 25 marks

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6. Ms Janis Smith takes out an endowment policy with an insurance company which involves making a fixed payment of \$P each year. At the end of n years, Janis expects to receive a payout of a sum of money which is equal to her total payments together with interest added at the rate of α % per annum of the total sum in the fund.

- (a) Show that the total sum in the fund at the end of the second year is $\$P(R + R^2)$, where

$$R = \left(1 + \frac{\alpha}{100}\right). \quad [7 \text{ marks}]$$

- (b) Show, by mathematical induction or otherwise, that the total sum in the fund at the end of the n^{th} year is

$$\$ \frac{PR(R^n - 1)}{R - 1}.$$

[12 marks]

- (c) Find the value of P, to the NEAREST dollar, when $n = 10$, $\alpha = 8$ and the payout is \$100,000.00.

[6 marks]

Total 25 marks

END OF TEST